# Leonzilla

version 1.11.6

Uploads your projects from the PC to the web with one click – over FTP, FTPS or SFTP. It also builds, backs up to ZIP and pushes to GitHub. It never deletes anything on the server.

- passwords protected by a master password
- portable – nothing to install
- only uploads, never deletes

This guide is also built into the app – in Leonzilla click the **?** button in the top bar (next to the settings) or press **F1**.

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## Quick start

1. **New card** – the `+` button in the bar or `Ctrl`+`T`. Every project (website, firmware...) has its own card.
2. **Fill in what the card should do** – any of these is enough: *synchronization to a server* (server + folder pairs PC → server), *build*, *backup*, *GitHub*, *FW build*. Everything is saved immediately. The *Synchronize* button appears once both a server and a folder are filled in; a backup-only card leaves the folder section empty.
3. **Done – lock.** A locked card only shows its settings and cannot be changed by accident. Change it with *Edit card*.
4. **Overview** – all cards as tiles, sorted into groups (Settings → Groups). Drag a tile to move it to another group. Click a group name to collapse it – each computer remembers its own collapsed groups (and open tabs).

5 **Tabs at the top** = open cards only. Open a card by clicking its tile, close it with ✕ on the tab (or `Ctrl` + `W`, middle mouse button) – it stays in the Overview. When the tabs do not fit, arrows ‹ › appear (the mouse wheel scrolls too).

A similar project? **Duplicate** copies the whole card including the password – then just change what differs. Delete a card you no longer need in *Edit card → Delete* or with a right click on its tab or tile (files on the PC and on the server are not touched).

## Buttons

**↑ Synchronize**

Compares the folders with the server and uploads new and changed files. You confirm before it starts (and tick the checklist items, if any are set).

**Build + Synchronize**

Compiles the selected builds first and uploads only when all of them succeed. *Build only* / *Upload only* does just one part.

**Check**

A preview without uploading – shows what would be uploaded.

**Back up**

Packs the configured folders into ZIP files with the date and time in the name.

**GitHub**

Goes through the card's repositories, shows the changes and commits and pushes.

**Duplicate**

A new card with the same settings, right after the original.

## Exclusions and patterns

What should not be uploaded or backed up. Exclusions can be set for each folder pair / backup (buttons *folder*, *file*, *pattern*) and for the whole card in *Exclude in all folders* (or *in all backups*) – one rule per line.

| PATTERN | EXCLUDES | EXAMPLE |
|---|---|---|
| `.git/` | a folder with this name **anywhere** | `.git/` `managed_components/` |
| `build*/` | folders whose name **starts** with build – anywhere | build, build_dev, build_prod |

| PATTERN | EXCLUDES | EXAMPLE |
|---|---|---|
| `*.psd` | files by mask – anywhere | `*.tmp` `~$*` |
| `img/raw/` | one **specific** folder (a slash in the middle = path from the source folder) | only img/raw, not other raw |
| `firmware/esp32/build*/` | a mask in the given path only | build folders only in firmware/esp32 |
| `docs/manual.pdf` | one specific file | |
| `photo?.jpg` | `?` = exactly one character | photo1.jpg, photoA.jpg |
| `# note` | a line starting with # is ignored | |

- **A trailing slash** = folders only (a file with the same name stays).
- `*` does not cross a slash: `a/*/` matches `a/x/`, not `a/x/y/`. A pattern without a slash in the middle applies on every level.
- Letter case does not matter, `\` and `/` mean the same.
- An excluded folder is not walked at all – this makes the comparison faster.

## What gets uploaded

- The file is **missing** on the server or has a **different size** → it is uploaded.
- The file on the PC **changed since the last upload** (Leonzilla remembers the size and time) → it is uploaded even with the same size.
- **Nothing is deleted** on the server – files deleted on the PC stay there.
- Files are uploaded through a temporary file that is renamed when complete (the website never sees a half-uploaded file), and the size is verified.
- To be sure, tick *Upload all files again* in the confirmation.

## Build (RAD Studio / TMS WEB Core)

- Builds are created once in *App settings* → *Builds* (a `.dproj` project, configuration, switches); a card just selects and orders them.
- Switches are read from the top part of `Predefines.inc` in a `{$IFNDEF LEONZILLA} … {$ENDIF}` block: RAD Studio uses the file as it is, a Leonzilla build skips the block and passes `LEONZILLA` + the switches ticked in the build. When a new one appears, load it with *Load switches*.
- Optionally line endings are converted to CRLF and the project's post-build event runs. The output (TMSWeb) is excluded from Dropbox so that Dropbox does not lock the files.

- **Version info** (in the build settings, 1 line, max 80 characters, default = build name): the post-build and the command after build get it in the `LEONZILLA_VERSION_INFO` variable – e.g. to write it into `version.json`. If your project's script reads a different variable name, add it in *App settings → Builds*. Empty = no info. The text can be changed in the build confirmation – **the card remembers it** and offers it next time; ↺ restores the default from the build settings.
- The progress window shows the steps and the compiler output; *Hide* keeps the build running in the background.
- **Build light** (a section in the card settings): instead of a RAD Studio project just a **path to a script** – Python (`.py`), batch (`.bat`, `.cmd`) or PowerShell (`.ps1`) – and optional parameters. The script runs in its own folder (Python from PATH, otherwise the `py` launcher); when it ends without an error (exit code 0), the card uploads the folders to the server. On the card it looks and works like a build – *Build + Synchronize*, *Build only*, *Upload only*. When the card also has RAD Studio builds, Build light runs after them. An own **button label** (e.g. *Build and upload to the web*) replaces the default text – handy for a script that uploads by itself (such as publishing a program on a website); such a card needs no server or folders.

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## Backups

- Each backup = folder → target folder. It creates `prefix_YYYY-MM-DD_hh-mm-ss.zip` (empty prefix = folder name).
- ZIP name in the confirmation – for each backup separately, with the *Note* | *Own name* switch: empty field = default `web_2026-10-18_10-41-14.zip` · *Note* = inserted between the name and the date `web_new cfg_2026-10-18_10-41-14.zip` · *Own name* = exactly your name without a date `fw-160.zip` (if it exists, `fw-160 (2).zip` is created – nothing is overwritten). You see the resulting name immediately; `Enter` jumps to the next field, in the last one it starts the backup.
- When a card has more backups, each one has a **checkbox** in the confirmation (ticked by default). An unticked backup is skipped this time – the card settings do not change.
- The name fields can be turned off for the whole card: *Edit card → Backups → Offer a note or own ZIP name when backing up*. The confirmation then only shows the default ZIP names.
- Exclusions work as with synchronization – typically `.git/`, `build*/`, `managed_components/`.
- The ZIP is written as `.zip.part` and renamed only when complete. A locked file is skipped with a warning. Leonzilla never deletes old backups.

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## GitHub

- Select the folders with git repositories in the card. Each repository knows where it belongs (`origin`).
- The GitHub button goes through them one by one and **skips those without changes**. Otherwise it shows the changed files (click = changes in the file), a commit message and the buttons *Commit + Push*, *Commit only*, *Skip*.

- **Only ticked files** are committed. Push goes to the current branch (usually `main`).
- *Pull changes before push* – when you also commit from another PC.
- Needs **Git for Windows**. Git handles the sign-in itself – Leonzilla stores no token.

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## FW build and monitor (ESP-IDF / ESP32)

- **Monitor** (a checkbox in the *FW build and monitor* section): the card and its tile get a *Monitor* button – pick a COM port, *Open monitor* and it runs, even without a build or flashing. The monitor alone needs no project folder.
- In the card's *FW build* section: the project folder (with `CMakeLists.txt`), the ESP-IDF activation (PowerShell profile from EIM – empty = found in `C:\Espressif\tools`) and **variants**. The *Load from tools\build_*.cmd* button creates them (name, build folder from `-B`, PROD scripts as "no flashing").
- The **FW build** button: pick a variant, an action (*Build*, *Build + Flash*, *Build + Flash + Monitor*, *Monitor only*) and a COM port. The choice is remembered per card.
- **Clean config** is ticked at start time (every time again) – it deletes `<build folder>\sdkconfig`, which is then recreated from `sdkconfig.defaults`. Needed after changing `sdkconfig.defaults`, the profile or Kconfig.
- **Reconfigure** (ticked by default, again each time the window opens): runs `tools\build_dev.cmd reconfigure` before the build (reloads CMake / sdkconfig).
- The **monitor** is right in the window: a coloured device output (errors red, warnings yellow, `@…` replies blue) and a command line at the bottom – `Enter` sends, `↑` `↓` history, *Device reset* = restart via RTS. Select text in the output with the mouse and copy it with `Ctrl`+`C`; top right *Copy all* and *Clear*. The output does not scroll while you select. *Hide* keeps the monitor running (the tile shows "FW monitor running").
- *ESP-IDF terminal* opens a normal PowerShell with ESP-IDF active in the project folder – for `menuconfig` and other commands.
- No other program (e.g. a VS Code monitor) may have the port open – Leonzilla tells you.

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## Passwords and security

### Without a master password

Server passwords in `config.json` are **obfuscated** (AES with a key inside the app). They are not readable in the file, but whoever has the file and Leonzilla can get them.

### With a master password

Passwords are **encrypted** – the key is derived from the master password (scrypt) and encrypted with AES-256-GCM. Without the master password nobody can read them from `config.json`. Turn it on in *App settings*; Leonzilla asks for it at start.

- **A forgotten master password cannot be recovered** – you then enter the server passwords again.
- SFTP: the server key is remembered on the first connection; when it changes, Leonzilla refuses to connect and asks.
- The window talks only to Leonzilla on this PC (`127.0.0.1`, a random port and token). Nothing is sent anywhere else.

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## Portable and more PCs

- Nothing to install: `Leonzilla.exe` and the `Leonzilla_data` folder next to it (cards, settings, upload state, logs). All of it can live in Dropbox and run on several PCs.
- Only the window size and position and the window cache are kept per PC (outside Dropbox, in `%LOCALAPPDATA%\Leonzilla`).
- Changes from another PC are loaded when you return to the window. Do not synchronize the same card on two PCs at the same time.
- One Leonzilla runs per PC – starting it again switches to the open window.
- **Start with Windows** (App settings → Start, off by default): Leonzilla starts minimized in the taskbar after you sign in. Set per PC.
- Needs Windows 10/11 with WebView2 (always present in Windows 11).
- **Language** – Czech or English: *App settings* → *Language* (default by the Windows display language).

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## Keys and tricks

**Ctrl** + **T** new card

**Ctrl** + **Tab** next card (**Shift** = previous)

**Ctrl** + **Enter**
in a commit message = Commit + Push

**Esc** closes a window / dialog

**Ctrl** + **W**
closes the card tab (the card stays in the Overview)

**right click**
on a tab or tile: edit, duplicate, colour, delete (the colour can also be changed at the top of the card editor)

**double click**
on the tab of a card being edited = rename

- Tabs and tiles can be dragged (order, groups).
- **Click the logo** top left = *About* (version, links to GitHub and the latest version, bug reports).

- The *Log* button opens the card's full log; after a crash the output is in `Leonzilla_data\logs\crash.log`.



## Thank you for using Leonzilla.

## Enjoy it!

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☕ Support development – buy the author a coffee